

phase_03_GAP_AUDIT

Phase 3 Gap Audit — Code-Grounded

Field	Value
Date	2026-06-01
Auditor	Engineering auditor (code-grounded, anti-bluff per CLAUDE-1 / §7.1)
Scope	docs/research/ PHASE_3_ROADMAP.md deliverables vs. actual pkg/ internal/ cmd/ api/
Honest completion	~70% of Phase 3 P0/P1 scope DONE; ~15% PARTIAL; ~15% MISSING/ DEFERRED

3-Axis Package Status (Refreshed 2026-06-04, HXC-939)

Why this section exists: The narrative table below mixes “exists” with “used”. A package can be **implemented** and **tested** yet never reached from any binary (**orphaned**). **wired** is **measured** via `go list -deps ./cmd/... | grep -Fx <module-path>/<pkg> (module = github.com/HelixDevelopment/helix_cluster)`, not assumed. **“Completed (registry) ≠ wired”** — Completed in the HXC registry means source+tests exist; it does NOT prove a shipped binary reaches it.

Package	implemented	wired (reachable from cmd/)	tested
pkg/etcd	yes	yes	yes
pkg/lock	yes	NO (orphaned)	yes
pkg/leader	yes	NO (orphaned)	yes
pkg/scheduler	yes	yes	yes
pkg/classads	yes	yes	yes

Package	implemented	wired (reachable from cmd/)	tested
internal/scheduler	yes	yes	yes
internal/node	yes	yes	yes
pkg/discovery	yes	yes	yes
pkg/security	yes	yes	yes
internal/security	yes	yes	yes
internal/gateway	yes	yes	yes
internal/policy	yes	yes	yes
pkg/metrics	yes	yes	yes
pkg/tracing	yes	NO (orphaned)	yes
internal/health	yes	yes	yes
pkg/health	yes	yes	yes
pkg/storage	yes	NO (orphaned)	yes
pkg/build	yes	yes	yes
internal/build	yes	yes	yes
internal/gpu	yes	yes	yes
pkg/gateway (roadmap name)	NO (only internal /gateway exists)	n/a	n/a

Orphan callouts (implemented+tested but NOT reachable from any cmd/binary): pkg/lock, pkg/leader, pkg/tracing, pkg/storage. The audit text already flags pkg/discovery’s etcd backend and pkg/classads as “implemented but unwired into the cluster/scheduler path” — the package-level wired column above is the mechanical version of that finding (note: the *package* pkg/classads is compiled into a binary, but its scheduler-Filter/Score integration is still absent, so feature-wiring ≠ package-wiring here).

One-line summary: The P0 consensus/scheduler/security/gateway skeletons and most cmd binaries are genuinely implemented with real-behavior tests (real etcd/ NATS/Postgres via brokertest), but several integrations are unwired (etcd discovery backend, ClassAds→scheduler, OTel tracing) and the GPU/storage/migration backends remain stubs or pure-Go placeholders.

Deliverable Status Table

Evidence paths are repo-relative. “DONE” = implemented AND has real-behavior tests; “PARTIAL” = implemented but unwired/mock-only/missing a claimed capability; “MISSING/DEFERRED” = stub/absent.

Deliverable (roadmap)	Status	Evidence file:line
pkg/etcd client wrapper	DONE	pkg/etcd/package.go:9,25 (real clientv3); pkg/etcd/etcd_integration.go:build integration)
pkg/lock distributed locks	DONE	pkg/lock/lock.go:60–74 (EtcdLocker on concurrency); pkg/lock_integration_test.go:27 (real etcd TestMain)
pkg/leader / raft-ish election	PARTIAL	pkg/leader/*.go (385 LoC, 411 test)
pkg/ scheduler Omega core + queue + preempt	DONE	pkg/scheduler/scheduler.go:8–50, gang_preempt.go, cost_gang_test.go (1129 test LoC)
pkg/classads expression parser/ evaluator	DONE (standalone)	pkg/classads/parser.go, ast.go, eval.go (662 LoC, 627 test)
internal/ scheduler gRPC service	PARTIAL	internal/scheduler/server.go:41,102 (wired to pkg/scheduler)
internal/ node agent (SWIM + resources + register)	PARTIAL	internal/node/node.go:79 (real SWIM), :11 (pkg/resources pr
pkg/ discovery etcd backend	PARTIAL	pkg/discovery/etcd_backend.go + etcd_backend_test.go (rea
pkg/security mTLS + SPIFFE	DONE	internal/security/orchestrator.go:136–164 (real rsa/x509 ce trust domain); pkg/security/tls.go:33; security_test.go (120
internal/ security RBAC/authz	DONE	internal/security/authorize_rbac_test.go, scopes_test.go
e2ee / attestation / gputest	DONE (submodule)	security/pkg/e2ee, security/pkg/attestation, security/pkg

Deliverable (roadmap)	Status	Evidence file:line
pkg/gateway / internal/gateway reverse proxy	PARTIAL	internal/gateway/gateway.go:24-33 (static route table, X-Helix gateway_test.go
internal/policy OPA engine	PARTIAL	internal/policy/engine.go (243 LoC), engine_test.go
pkg/metrics Prometheus exposition	DONE	pkg/metrics/package.go:64 (Histogram), :34 (Gauge), :172 (Prometheus)
pkg/tracing OpenTelemetry	PARTIAL	pkg/tracing/package.go:21, 47 (homegrown TraceID/Span)
internal/health monitor + scoring	DONE	internal/health/*.go (592 LoC, 802 test, 4 test files); pkg/health
pkg/storage abstraction	PARTIAL	pkg/storage/storage.go:34 (Memory), :93 (File)
pkg/build RBE/CAS	DONE	pkg/build/*.go (574 LoC), build_integration_test.go; internal
cmd/helixd control daemon	PARTIAL	cmd/helixd/main.go:180, 245 (dependency health prober + HTTP
cmd/htmux CLI	DONE	cmd/htmux/main.go:97-105 (create/attach/list/kill/rename), client sessionClient)
All other cmd/* binaries	DONE	cmd/helix-{scheduler,session,security,gateway,node,build,health,pkg} main.go (each 140-420 LoC + a test file)
internal/gpu/* CUDA/ROCm/oneAPI/MLX backends	DEFERRED	internal/gpu/manager.go, gpu.go (640 LoC, pure Go; grep "im
Session live migration	DEFERRED	pkg/session/migration.go:178-222 ("criu not implemented"

Deliverable (roadmap)	Status	Evidence file:line
(CRIU/DMTCP)		
WireGuard NAT traversal (UPnP/NAT-PMP)	DEFERRED	pkg/wireguard/nat_traversal.go:35,41 ("UPnP/NAT-PMP not
True Raft consensus (pkg/raft)	MISSING	(no pkg/raft)

TOP IMPLEMENTABLE GAPS (Go, no new infra/hardware)

Prioritized. Each is buildable with existing deps and produces an end-user-visible behavior change with a mutation-pairable acceptance test.

1. Wire EtcdBackend into the node agent (real cluster-wide discovery)

- **Target dir:** internal/node/
- **Spec:** internal/node/node.go:105 hardcodes `discovery.NewInMemoryBackend()`, so node registrations are invisible cluster-wide despite a complete, tested `pkg/discovery/etcd_backend.go`. Add an Agent config field (`EtcdEndpoints []string`); when non-empty, construct `discovery.NewEtcdBackend(...)` with a TTL lease tied to the SWIM heartbeat so a dead node's instance auto-expires. Fall back to in-memory only when no endpoints are configured. Expose the chosen backend type via the agent's status so operators can see "etcd" vs "memory".
- **Acceptance test (mutation-pairable):** Integration test (`//go:build integration`) boots a real etcd via `brokertest`, starts two agents pointing at it, and asserts agent A's `registry.List()` returns agent B's instance with B's resource metadata. **Mutation:** revert `node.go` to `NewInMemoryBackend()` → test must FAIL (A sees only itself).

2. Make internal/scheduler.StreamJobEvents a real event stream

- **Target dir:** internal/scheduler/ (+ small `pkg/scheduler` event hook)
- **Spec:** `server.go:193` sends a single hardcoded "scheduled" event and returns — a PASS-bluff for "StreamJobEvents". Add a per-job event channel in `pkg/scheduler` (publish on SCHEDULED / BOUND / PREEMPTED / COMPLETED / CANCELLED state transitions) and have the gRPC handler subscribe and forward events until job terminal or client cancel.
- **Acceptance test:** Schedule a job, open the stream, then cancel/preempt it; assert the stream emits ≥ 2 distinct `EventTypes` in order with monotonically increasing

timestamps. **Mutation:** drop the PREEMPTED publish → test must FAIL (stream missing the second event).

3. Connect pkg/classads to the scheduler Filter/Score stages

- **Target dir:** pkg/scheduler/ (new plugins_classads.go)
- **Spec:** A full ClassAds parser/evaluator exists (pkg/classads/eval.go) but `grep classads pkg/scheduler` returns nothing — the scheduler ignores it. Add a `ClassAdFilterPlugin` and `ClassAdScorePlugin` implementing the existing `Plugin` interface: evaluate a job's `Requirements` expression against each node's attribute ad (Filter), and a Rank expression to produce a Score. Reuse `pkg/scheduler.Resources` as the attribute source.
- **Acceptance test:** Build two nodes (one matching, one not) and a job whose `Requirements = (GPU >= 1)`; assert the filter rejects the non-GPU node and `Rank = -Memory` orders nodes by free memory. **Mutation:** flip the comparison operator in the evaluator call (`>=><`) → test must FAIL.

4. Add auth + RBAC enforcement to the gateway

- **Target dir:** internal/gateway/
- **Spec:** `gateway.go` is a bare reverse proxy with zero auth (`grep tls|auth|jwt` → none), contradicting roadmap §2.7 (mTLS + OPA enforcement). Add a middleware that extracts a bearer/JWT identity (reuse `pkg/jwt`) and checks required scopes (reuse `internal/security scope model / internal/policy`) per route prefix; reject with 401/403 and an `X-Helix-Deny-Reason` header before proxying.
- **Acceptance test:** Request `/api/v1/scheduler/` with (a) no token → 401, (b) token missing `scheduler:write` → 403 with deny reason, (c) valid token → proxied (sees `X-Helix-Gateway`). **Mutation:** make the middleware fail-open on missing scope → cases (a)/(b) must FAIL.

5. Add an S3-compatible (minio) backend to pkg/storage

- **Target dir:** pkg/storage/
- **Spec:** Only Memory + File exist; roadmap §2.4 wants pluggable distributed backends. Add an `S3Store` implementing the existing `Put/Get/Delete/List` interface against the standard S3 API (minio-go or aws-sdk-go-v2). No new infra at test time — the minio container is already in the compose stack, so the integration test can boot it via the existing `brokertest` harness pattern.
- **Acceptance test (//go:build integration):** Round-trip a blob through `S3Store` against a real minio container; assert `Get` returns bytes-identical content and `List(prefix)` enumerates it. **Mutation:** swap `Put` to write to the wrong bucket → `Get` must FAIL.

6. OTLP/Jaeger export shim for pkg/tracing

- **Target dir:** pkg/tracing/
- **Spec:** Tracing is homegrown with no exporter, so spans never reach the deployed Jaeger. Add a thin OTLP-HTTP exporter that serializes existing `Spans` (`TraceID/ParentID/timings/attrs`) to the OTLP JSON span shape and POSTs batches to the collector endpoint. Keep the homegrown API; only add a sink. (Stops short of full OTel SDK migration, which is larger.)

- **Acceptance test:** Stand up an httptest. Server as a fake collector; emit a parent+child span; assert the exporter POSTs one batch containing both spans with the correct parent linkage and a shared TraceID. **Mutation:** drop ParentID from the serialization → linkage assertion must FAIL.

DEFERRED (need external infra / hardware / Zig / GPU kernels) — count: 4

Item	Reason deferred
internal/gpu/* CUDA/ROCm/oneAPI/MLX backends	Requires physical GPUs + vendor SDKs (CUDA/ROCm/oneAPI/MLX) and cgo bindings; cannot be validated end-user-usable without hardware (CLAUDE-1 sink-side evidence impossible in CI).
Session live migration (CRIU/DMTCP)	Requires CRIU/DMTCP host binaries and privileged kernel checkpoint support; explicit stubs at pkg/session/migration.go:178–222.
WireGuard UPnP/NAT-PMP traversal	Requires a real NAT gateway/router to exercise; cannot prove traversal in a flat CI network.
True embedded Raft (pkg/raft)	Implementable in Go (hashicorp/raft), but a multi-week consensus deliverable, not a quick gap; current etcd-backed locks/election cover the immediate need, so this is deferred as a major work item rather than a quick win.